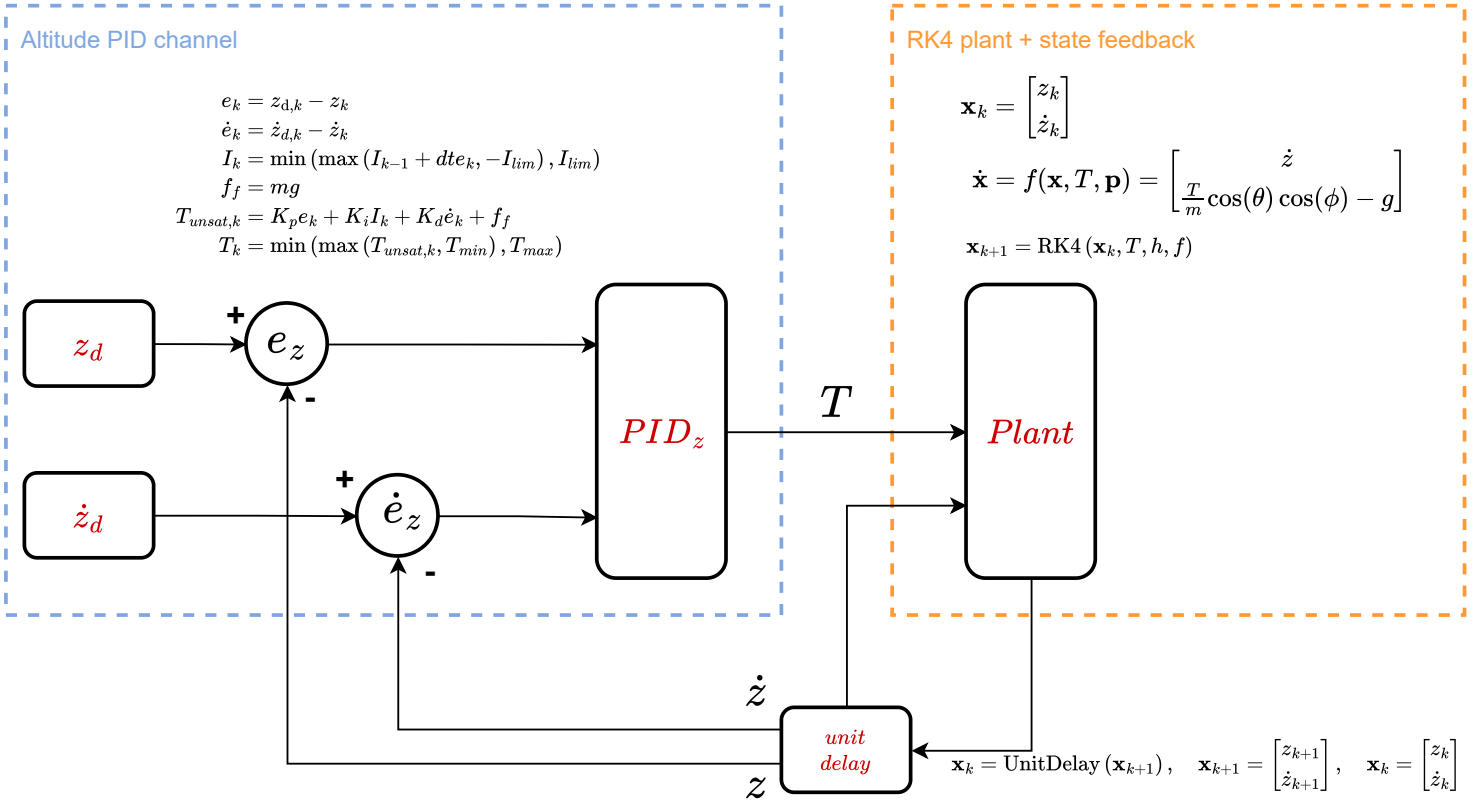

Altitude PID channel
$e_k = z_{d,k} - z_k$
$\dot{e}_k = \dot{z}_{d,k} - \dot{z}_k$
$I_k = \min\left(\max\left(I_{k-1} + dte_k, -I_{lim}\right), I_{lim}\right)$
$f_f = mg$
$T_{unsat,k} = K_p e_k + K_i I_k + K_d \dot{e}_k + f_f$
$T_k = \min\left(\max\left(T_{unsat,k}, T_{min}\right), T_{max}\right)$
RK4 plant + state feedback
$\mathbf{x}_k = \begin{bmatrix} z_k \\ \dot{z}_k \end{bmatrix}$
$\dot{\mathbf{x}} = f(\mathbf{x}, T, \mathbf{p}) = \begin{bmatrix} \dot{z} \\ \frac{T}{m}\cos(\theta)\cos(\phi) - g \end{bmatrix}$
$\mathbf{x}_{k+1} = \mathrm{RK4}\left(\mathbf{x}_k, T, h, f\right)$
z_d
e_z
+
-
$\dot{z}_d$
$\dot{e}_z$
+
-
PID_z
T
$Plant$
$\dot{z}$
z
unit delay
$\mathbf{x}_k = \mathrm{UnitDelay}\left(\mathbf{x}_{k+1}\right), \quad \mathbf{x}_{k+1} = \begin{bmatrix} z_{k+1} \\ \dot{z}_{k+1} \end{bmatrix}, \quad \mathbf{x}_k = \begin{bmatrix} z_k \\ \dot{z}_k \end{bmatrix}$